Foundations of Generative AI (ICT4403)

Assignment 2 — Submission Details

Student Name	NehalVerma
Registration Number	230953144
Course	Foundations of Generative AI (ICT4403)
Assignment Title	Assignment 2 – Autoregressive Text Generation using Recurrent Neural Networks
Title of the Developed System	Autoregressive Next-Word Text Generation System with a Comparative Evaluation of Simple RNN, Bidirectional RNN, LSTM and GRU Models
Dataset Used	Alice's Adventures in Wonderland by Lewis Carroll — Project Gutenberg eBook #11
Dataset Link	https://www.gutenberg.org/ebooks/11
Dataset File (plain text)	https://www.gutenberg.org/cache/epub/11/pg11.txt
GitHub Repository Link	https://github.com/Nehalll-code/GENAI_PROJECT

Summary of the Developed System

The system is a word-level autoregressive language model that predicts the next word from a fixed 20-word context window and generates text by feeding each sampled word back into that window. Four recurrent architectures — Simple RNN, Bidirectional RNN, LSTM and GRU — were trained under identical conditions on the same corpus and compared quantitatively (cross-entropy loss, perplexity, next-word accuracy and training time) and qualitatively (coherence, diversity, contextual consistency and repetition of the generated text).

Text was then generated from three fixed seed sequences at sampling temperatures of 0.5, 1.0 and 1.5 for every model, and the effect of temperature on coherence, diversity and repetition was analysed. The repository contains the full implementation, the executed Jupyter notebook, the metric tables, the comparison plots and the generated-text samples.